

KOUROSH GHOLAMI, EIT, PhD

Phone: (825) 558-8595, Email: kgholami@ualberta.ca, [LinkedIn](#)

Professional Summary

- Engineer-in-Training (EIT) with a PhD focus in Mining Engineering and strong background in geotechnical and civil engineering, experienced in energy, infrastructure, and materials.
- 7+ years of experience in advanced numerical modeling and optimization techniques, improving system performance and computational efficiency.
- 6+ years of hands-on laboratory and experimental experience in concrete/material testing, physical modeling, and development of low-carbon, high-performance materials.
- 6+ years of professional experience in teamwork, project coordination, client communication, and conflict resolution, developed in a consulting company and through industry collaborations.

Education

- **PhD Candidate — Civil Engineering — Mining Engineering**
University of Alberta, Edmonton, Canada | 2022 – Present
- **MSc — Civil Engineering — Geotechnical Engineering**
University of Tehran, Iran | 2019 – 2022
- **BSc — Civil Engineering**
University of Tehran, Iran | 2015 – 2019

Engineering Experience

Research Assistant (PhD – Mining Engineering)

University of Alberta | 2022 – Present

- Built optimization framework for geothermal energy piles to improve efficiency using FEM & Genetic algorithm, reducing computational cost by >95%.
- Developed freeze-thaw-resistant cementitious materials using phase change materials, improving freeze-thaw performance and thermal efficiency (COP) by 7%.
- Engineered low-carbon backfill materials using LC³, reducing carbon emissions by 33%.
- Designed low-carbon thermal insulation concrete using local wood chips and BCSA cement, achieving up to 7× improvement in early-age UCS and proper insulation.
- Conducted advanced simulations using COMSOL for coupled thermomechanical systems.

Research Assistant (MSc – Geotechnical Engineering)

University of Tehran | 2019 – 2022

- Designed and analyzed wave barrier systems to reduce ground vibration, achieving up to 40% reduction in wave transmission.
- Applied optimization algorithms to reduce computational cost by over 70%.
- Conducted numerical studies on soil behavior in saturated and dry conditions.

Tender & Cost Estimation Engineer

Shora Consultant Engineers Co. | 2016 – 2019

- 3+ years of experience in client communication, project coordination, and conflict resolution within a consulting environment, supporting multidisciplinary engineering projects.
- Prepared detailed cost estimates and tender documents for civil engineering projects.
- Supported bid preparation, contract documentation, and project evaluation processes.

- Contributed to project planning, budgeting, and feasibility assessments.

Intern & Resident Engineer — Baam Engineering Development Group | 2020

- Supported on-site engineering activities and assisted with project coordination and execution.

Selected Publications

- Gholami et al. “Optimizing the layout of geothermal energy piles to minimize ground temperature changes.” *Construction and Building Materials* (2025).
- Gholami et al. “Developing thermal insulation concrete with enhanced mechanical strength using belitic calcium sulfoaluminate cement and wood chips.” *Construction and Building Materials* (2024).
- Gholami et al. “Shear and compression waves screening in 2D for dry or saturated ground using periodic infinite and finite soil-foam barriers.” *Archive of Applied Mechanics* (2023).

Public Presentations

- A data-driven intelligent approach for designing sustainable cementitious mixtures with mine tailings — CIM Connect 2026, Vancouver, Canada
- Presented research on geothermal energy piles and thermomechanical modeling in academic and industry meetings, University of Alberta (2022–Present)
- Wave barrier design using solid-state physics concepts — 13th National Congress on Civil Engineering, Isfahan, Iran (2022)

Honors & Awards

- Alberta Innovates Graduate Student Scholarship (\$31,000/year, 2024–2026)
- Top 20% — BSc & MSc, University of Tehran

Leadership and Volunteering Experience

- Organized and led the annual Civil Engineering department ceremony at the University of Tehran, hosting ~2,000 attendees (faculty, students, and industry professionals); managing the event, sponsorship acquisition, team coordination, budgeting, permitting, and guest invitations.
- Volunteered in post-earthquake damage assessment for the Sar-Pol-Zahab earthquake, supporting building evaluation and recovery efforts.

Core Skills

Numerical Simulation & Software:

- Finite Element Analysis (COMSOL, ABAQUS)
- Structural & Mining Tools: ETABS, SAP2000, Deswik

Programming & Optimization:

- MATLAB, Python
- Optimization: Genetic Algorithms, NSGA-II

Laboratory & Experimental Skills:

- Concrete testing (UCS, material characterization)
- Thermal analysis (thermal conductivity, TGA)
- Physical modeling (sandbox, shaking table)

Professional Skills:

- Teamwork, project coordination, client communication and stakeholder engagement, conflict resolution, problem-solving, verbal and written communication